

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 8: Databases

Subtopic 8.2: Database Management Systems (DBMS)

Concept 8.2.1: DBMS Features

Total Questions: 13

Mark Schemes begin on Page 8

Question 1 | Source: 9618_w25_qp_12 (Q4.d)

- 4 A relational database, SHIPPING, stores data about the ships in a company and the containers that are carried on the ships.

The database has the following tables:

CONTAINER(ContainerID, Type, Weight, OwnerName, ShipID)

SHIP(ShipID, Type, Capacity, ShipName)

- (d) Describe the purpose of a developer interface in a Database Management System (DBMS).

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..... [2]

Question 3 | Source: 9618_s25_qp_13 (Q6.d.i)

- 6 A shop sells pens to customers. Customers place an order with the shop and collect the items the next day. The shop uses a database to store the information about the orders.

The database contains the following tables:

CUSTOMER(CustomerID, CustomerName, Email)

ORDER(OrderID, CustomerID, Date, Collected)

ORDER_PRODUCT(OrderID, ProductID, Quantity)

PRODUCT(ProductID, ProductName, QuantityInBox, Cost, SupplierID)

SUPPLIER(SupplierID, SupplierName, SupplierEmail)

The primary keys are underlined.

- (d) The shop uses a Database Management System (DBMS).

- (i) One feature provided by a DBMS is a data dictionary.

The data dictionary stores the attribute names, table names, foreign keys and primary keys in the database.

Identify **three** other items stored in a data dictionary.

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3 [3]

Question 4 | Source: 9618_w24_qp_13 (Q4.e)

4 A company makes ice cream and sells it to shops.

The ice cream is made in batches: a large quantity of one type and flavour of ice cream that is then split into smaller quantities for sale.

The company’s owner has designed a relational database, ICECREAM, to store data about their ice cream and customers.

Some of the tables in the database are given. The database is not normalised.

BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)

CUSTOMER(CustomerID, CompanyName, EmailAddress, TelephoneNumber)

SALE(SaleID, BatchID, CustomerID, Quantity, Date)

(e) A Database Management System (DBMS) supports data integrity.

Explain how a DBMS supports data integrity.

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..... [3]

Question 5 | Source: 9618_s24_qp_11 (Q6.b)

6 A company is developing a website that will allow users to create an account and then play a quiz every day. The data about the users and the quizzes are stored in a database.

A user must select a unique username and enter a valid email address to create an account. All users must be over the age of 16. A new quiz is given to the users every day. Each quiz is stored in its own text file.

The database stores the filename of each quiz and the date it can be played. The user gets a score for each quiz they complete, which is stored in the database. The scores are used to give each user a rating, for example Gold.

- (b) The company is using a Database Management System (DBMS) to set up the database.

Describe what is meant by the following DBMS features:

Data dictionary

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Logical schema

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[4]

Question 6 | Source: 9618_s23_qp_11 (Q2.a.i)

- 2 An organisation uses a database to store data about the types of bird that people have seen.

- (a) The database is managed using a Database Management System (DBMS).

- (i) State what is meant by a data dictionary **and** give **one** example of an item typically found in a data dictionary.

Definition

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Example

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[2]

Question 7 | Source: 9618_s23_qp_11 (Q2.a.ii)

- 2 An organisation uses a database to store data about the types of bird that people have seen.

- (a) The database is managed using a Database Management System (DBMS).

- (ii) State what is meant by data integrity **and** give **one** example of how this is implemented in a database.

Definition

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Example

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[2]

Question 8 | Source: 9618_s23_qp_12 (Q2.a)

2 A horse riding school uses a database, Lessons, to store data about lesson bookings.
This database is created and managed using a Database Management System (DBMS).

(a) The table contains names and descriptions of DBMS features and tools.

Complete the table by writing down the missing names and descriptions.

Name	Description
Data dictionary	
Query processor	
.....	A model of a database that is not specific to one DBMS.
.....	A software tool that allows the user to create items such as tables, forms and reports.

[4]

Question 9 | Source: 9618_w22_qp_12 (Q5.d.i)

5 A relational database, GARDEN, has the following tables:

OWNER(OwnerID, FirstName, TelephoneNo, TreeID, TreePosition)

TREE(TreeID, ScientificName, MaxHeight, FastGrowing)

(d) (i) Describe, using an example, what is meant by a **data dictionary**.

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[2]

Question 10 | Source: 9618_w22_qp_12 (Q5.d.ii)

5 A relational database, GARDEN, has the following tables:

OWNER(OwnerID, FirstName, TelephoneNo, TreeID, TreePosition)

TREE(TreeID, ScientificName, MaxHeight, FastGrowing)

(ii) Describe what is meant by a **logical schema**.

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..... [2]

Question 11 | Source: 9618_w22_qp_13 (Q2.b)

2 The relational database ASTRONOMY is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

COMPANY(TelephoneNumber, CompanyID, CompanyName)

PHOTOGRAPH(PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)

TELESCOPE(TelescopeID, CompanyID, SerialNumber)

(b) A Database Management System (DBMS) has several features.

Identify the feature that describes the relationship between data and its structure.

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Question 12 | Source: 9618_w22_qp_13 (Q2.f)

2 The relational database ASTRONOMY is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

COMPANY(TelephoneNumber, CompanyID, CompanyName)

PHOTOGRAPH(PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)

TELESCOPE(TelescopeID, CompanyID, SerialNumber)

(f) Describe the purpose of a query processor in a DBMS.

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..... [2]

Question 13 | Source: 9618_w21_qp_12 (Q6.b)

6 A shop sells plants to customers. The shop manager has a relational database to keep track of the sales.

The database, PLANTSALES, has the following structure:

PLANT(PlantName, QuantityInStock, Cost)

CUSTOMER(CustomerID, FirstName, LastName, Address, Email)

PURCHASE(PurchaseID, CustomerID)

PURCHASE_ITEM(PurchaseID, PlantName, Quantity)

(b) The shop manager uses a Database Management System (DBMS).

Describe the purpose **and** contents of the data dictionary in the DBMS.

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..... [3]

Question 14 | Source: 9618_s21_qp_11 (Q7.c)

7 Bobby and Kim are discussing databases.

(c) A database has a data dictionary.

Give **three** items that are stored in a data dictionary.

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[3]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q4.d)

4(d)	1 mark per bullet point, max 2 marks • To allow the user to create / modify / delete tables // maintain the database • To allow the user to set up / modify relationships • To allow the user to create a form for data input • To allow a user to add tools to a form // for example, drop-down boxes / buttons etc. • To allow a user to design a report (to show the output in an organised manner) • To allow a user to add a menu to enable a choice of different actions • To allow a user to inspect the database contents and metadata • ... by creating and/or running different SQL queries	2
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Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q5.d)

5(d)	One mark for identification of method one mark for corresponding description max 4 marks • Authentication methods / passwords / biometrics / 2-factor authentication can be implemented • ... which prevents unauthorised access to the customer's data • Access rights / privileges can be set • ... so that only those with correct permissions can read / edit customer's data • Regular backups can be scheduled • ... so that a second copy of the customer's data is available in case of loss/damage • The data can be encrypted • ... so that the customer's data cannot be understood by anyone who gains unauthorized access • Different views can be created • ... so that not everyone can see the customer's data	4
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Mark Scheme for Question 3 | Source: 9618_s25_ms_13 (Q6.d.i)

6(d)(i)	1 mark each to max 3 e.g. - Relationships - Views - Data types - Validation rules	3
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Mark Scheme for Question 4 | Source: 9618_w24_ms_13 (Q4.e)

4(e)	1 mark for each bullet point each (max 3) - Referential integrity is enforced ... such as cascade update/delete // if the data is changed in one place it is updated in every other place ... and ensures each foreign key has a corresponding primary key	3
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Mark Scheme for Question 5 | Source: 9618_s24_ms_11 (Q6.b)

6(b)	1 mark each to max 2 for data dictionary and max 2 for logical schema: Data dictionary: - Data about the data in the database // metadata - Identifies the characteristics of the data that will be stored - Appropriate example e.g. field names, table name, validation rules, data types, primary / foreign keys, relationships etc. Logical schema: - Conceptual design - Platform/database independent overview of the database - Is used to design the physical structure - Appropriate example e.g. Design of entities / E-R diagram / views	4
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Mark Scheme for Question 6 | Source: 9618_s23_ms_11 (Q2.a.i)

2(a)(i)	1 mark for definition: - Data about the data in the database // data about the structure of the database // metadata for a database 1 mark for a suitable example Examples: - table names - data types - field names	2
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Mark Scheme for Question 7 | Source: 9618_s23_ms_11 (Q2.a.ii)

2(a)(ii)	1 mark for definition - Methods of making sure the data is consistent 1 mark for example Examples: - Enforcing referential integrity - If data in one table is deleted/edited all tables are updated // cascading update/delete - Validation/verification rules	2
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Mark Scheme for Question 8 | Source: 9618_s23_ms_12 (Q2.a)

2(a)

1 mark for each correct feature or description

4

Feature	Description
Data dictionary	Data about the data in the database // data about the structure of the database // metadata for a database
Query processor	Software that allows the user to enter criteria , then finds and returns the appropriate result // software that processes and executes queries written in SQL
<u>Logical schema</u>	A model of a database that is not specific to one DBMS
<u>Developer interface</u>	A software tool that allows the user to create items such as tables, forms and reports

Mark Scheme for Question 9 | Source: 9618_w22_ms_12 (Q5.d.i)

5(d)(i)	1 mark for description - stores metadata about the database 1 mark for a correct example For example: - field / attribute names - table name - validation rules - data types - primary keys // foreign keys - relationships	2
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Mark Scheme for Question 10 | Source: 9618_w22_ms_12 (Q5.d.ii)

5(d)(ii)	1 mark for each bullet point (max 2): • the overview of a database structure • models the problem / situation • ... by using methods such as an ER diagram • independent of any particular DBMS	2
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Mark Scheme for Question 11 | Source: 9618_w22_ms_13 (Q2.b)

2(b)	Logical schema	1
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Mark Scheme for Question 12 | Source: 9618_w22_ms_13 (Q2.f)

2(f)	1 mark for each bullet point (max 2): • allows the user to enter criteria • searches for the data that meets the entered criteria • organises the results to be displayed to the user	2
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Mark Scheme for Question 13 | Source: 9618_w21_ms_12 (Q6.b)

6(b)	1 mark for description of purpose • Stores metadata about the database 1 mark for each example of contents to max 2 e.g. • field / attribute names • table name • validation rules • data types • primary keys // foreign keys • relationships	3
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Mark Scheme for Question 14 | Source: 9618_s21_ms_11 (Q7.c)

7(c)	1 mark per item to max 3 • table name • field name // attribute • data type • type of validation • Primary Key • Foreign Key • relationships	3
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